

24BDS0339 ATIKSH CHAWLA

PLSQL

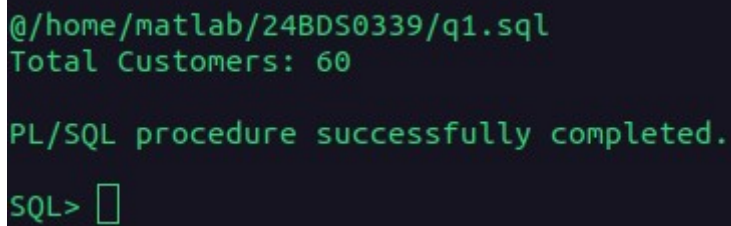
1) Display the total number of customers.

Code:

```
DECLARE
    n NUMBER;
BEGIN
    SELECT COUNT(*) INTO n
    FROM customer_0339;

    DBMS_OUTPUT.PUT_LINE('Total Customers: ' || n);
END;
/
```

Output:



```
@/home/matlab/24BDS0339/q1.sql
Total Customers: 60

PL/SQL procedure successfully completed.

SQL> █
```

2) Display the average product price.

Code:

```
DECLARE
    avg_price NUMBER;
BEGIN
    SELECT AVG(price) INTO avg_price
    FROM product_0339;

    DBMS_OUTPUT.PUT_LINE('Average Price: ' || avg_price);
END;
/
```

Output:

```
@/home/matlab/24BDS0339/q2.sql
Average Price: 22703

PL/SQL procedure successfully completed.

SQL> □
```

3) Accept a product ID and classify its price using case.

Classify products as:

< 5000 → Budget

5000–20000 → Mid-range

> 20000 → Premium

Code:

```
DECLARE
    pid NUMBER := &product_id;
    price NUMBER;
BEGIN
    SELECT price INTO price
    FROM product_0339
    WHERE product_id = pid;

    CASE
        WHEN price < 5000 THEN
            DBMS_OUTPUT.PUT_LINE('Budget');
        WHEN price <= 20000 THEN
            DBMS_OUTPUT.PUT_LINE('Mid-range');
        ELSE
            DBMS_OUTPUT.PUT_LINE('Premium');
    END CASE;
END;
/
```

Output:

```
@/home/matlab/24BDS0339/q3.sql
3067
old 2:      pid NUMBER := &product_id;
new 2:      pid NUMBER := 3067;
Premium

PL/SQL procedure successfully completed.

SQL> □
```

4) Accept a product ID and determine whether stock is low using if-else.

Rules:

0–10 → Low stock

11–50 → Moderate stock

>50 → Sufficient stock

Code:

```
DECLARE
    pid NUMBER := &product_id;
    stock NUMBER;
BEGIN
    SELECT stock_quantity INTO stock
    FROM product_0339
    WHERE product_id = pid;

    IF stock <= 10 THEN
        DBMS_OUTPUT.PUT_LINE('Low stock');
    ELSIF stock <= 50 THEN
        DBMS_OUTPUT.PUT_LINE('Moderate stock');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Sufficient stock');
    END IF;
END;
/
```

Output:

```
@/home/matlab/24BDS0339/q4.sql
3067
old 2:      pid NUMBER := &product_id;
new 2:      pid NUMBER := 3067;
Moderate stock

PL/SQL procedure successfully completed.

SQL> □
```

5) Display a multiplication table

Code:

```
DECLARE
    n NUMBER := &number;
BEGIN
    FOR i IN 1..10 LOOP
        DBMS_OUTPUT.PUT_LINE(n || ' x ' || i || ' = ' || n*i);
    END LOOP;
END;
/
```

Output:

```
@/home/matlab/24BDS0339/q5.sql
67
old 2:      n NUMBER := &number;
new 2:      n NUMBER := 67;
67 x 1 = 67
67 x 2 = 134
67 x 3 = 201
67 x 4 = 268
67 x 5 = 335
67 x 6 = 402
67 x 7 = 469
67 x 8 = 536
67 x 9 = 603
67 x 10 = 670

PL/SQL procedure successfully completed.

SQL> □
```

6) Display the following pattern:

```
*  
* *  
* * *  
* * * *  
* * * * *
```

Code:

```
BEGIN  
    FOR i IN 1..5 LOOP  
        FOR j IN 1..i LOOP  
            DBMS_OUTPUT.PUT('* ');  
        END LOOP;  
  
        DBMS_OUTPUT.PUT_LINE("");  
    END LOOP;  
END;  
/
```

Output:



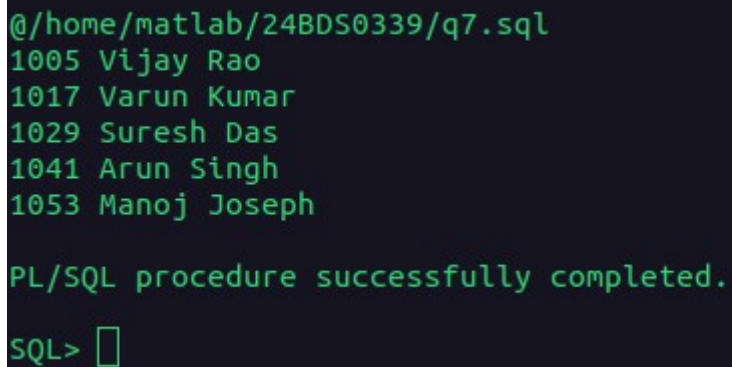
```
@/home/matlab/24BDS0339/q6.sql  
*  
* *  
* * *  
* * * *  
* * * * *  
  
PL/SQL procedure successfully completed.  
SQL> █
```

7) Using a cursor, display all customers from Chennai.

Code:

```
DECLARE
    CURSOR c IS
        SELECT customer_id, customer_name
        FROM customer_0339
        WHERE city = 'Chennai';
BEGIN
    FOR r IN c LOOP
        DBMS_OUTPUT.PUT_LINE(r.customer_id || ' ' || r.customer_name);
    END LOOP;
END;
/
```

Output:



The screenshot shows a terminal window with a dark background and green text. It displays the output of a PL/SQL cursor query, listing customer IDs and names for customers in Chennai. The output is as follows:

```
@/home/matlab/24BDS0339/q7.sql
1005 Vijay Rao
1017 Varun Kumar
1029 Suresh Das
1041 Arun Singh
1053 Manoj Joseph

PL/SQL procedure successfully completed.
SQL> 
```

8) Create a procedure to display a customer's order count.

Code:

```
CREATE OR REPLACE PROCEDURE order_count(cid NUMBER) IS
    n NUMBER;
BEGIN
    SELECT COUNT(*) INTO n
    FROM orders_0339
    WHERE customer_id = cid;

    DBMS_OUTPUT.PUT_LINE('Order Count: ' || n);
END;
/
```

Output:

```
@/home/matlab/24BD50339/q8.sql

Procedure created.

EXEC order_count(1067)
Order Count: 0

PL/SQL procedure successfully completed.

EXEC order_count(1077)
Order Count: 0

PL/SQL procedure successfully completed.

EXEC order_count(1041)
Order Count: 2

PL/SQL procedure successfully completed.

EXEC order_count(1043)
Order Count: 1

PL/SQL procedure successfully completed.

SQL> █
```


9) Create a function that returns the total spending of a customer.

Code:

```
RETURN NUMBER IS
    total NUMBER;
BEGIN
    SELECT NVL(SUM(oi.quantity * oi.unit_price), 0)
    INTO total
    FROM orders_0339 o
    JOIN order_item_0339 oi
    ON o.order_id = oi.order_id
    WHERE o.customer_id = cid;

    RETURN total;
END;
/
```

Output:

```
@/home/matlab/24BDS0339/q9.sql

Function created.

SELECT total_spending(101) AS total_spending
FROM dual;

@/home/matlab/24BDS0339/q9.sql

Function created.

SELECT total_spending(1043) From Dual;

TOTAL_SPENDING(1043)
-----
                589177

SELECT total_spending(1041) From Dual;

TOTAL_SPENDING(1041)
-----
                31785

SELECT total_spending(1067) From Dual;

TOTAL_SPENDING(1067)
-----
                 0

SQL> █
```

10) Create a trigger that displays the difference in the product price on update.

Code:

```
CREATE OR REPLACE TRIGGER price_change
BEFORE UPDATE OF price ON product_0339
FOR EACH ROW
BEGIN
    DBMS_OUTPUT.PUT_LINE('Price Difference: ' || (:NEW.price - :OLD.price));
END;
/
```

Output:

```
@/home/matlab/24BDS0339/q10.sql
Trigger created.
UPDATE product_0339 set price = price + 1000 where product_id = 3067;
Price Difference: 1000
1 row updated.
UPDATE product_0339 set price = price + 1000 where product_id = 3047;
Price Difference: 1000
1 row updated.
SQL> □
```

```
@24BDS0339/script3.sql
```

```
4
```

```
old 24:          x := &x;
```

```
new 24:          x := 4;
```

```
8
```

```
old 25:          y := &y;
```

```
new 25:          y := 8;
```

```
Max value is 8
```

```
Total employees is 11
```

```
PL/SQL procedure successfully completed.
```

```
SQL> 
```

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code:

```
DECLARE
    msg varchar2(20);
    x number(5);
    y number(5);
    z number(5);
begin
    msg := '&msg';
    dbms_output.put_line('Hello!' || msg);
    x:=&x;
    y:=&y;
    z:= x+y;
    dbms_output.put_line('Addition: ' || z);
end;
/
```

```
@/home/matlab/24BDS0339/script.sql
welcome
old   7:      msg := '&msg';
new   7:      msg := 'welcome';
1
old   9:      x:=&x;
new   9:      x:=1;
2
old  10:      y:=&y;
new  10:      y:=2;
Hello!welcome
Addition: 3

PL/SQL procedure successfully completed.

SQL> □
```

```

code:
DECLARE
    msg varchar2(20);
    a number(5);
    b number(5);
begin
    a:=&a;
    b:=&b;
    if(a>b) then
        dbms_output.put_line('A is greater');
    else
        dbms_output.put_line('B is greater');
    end if;
end;
/

```

```

@/home/matlab/24BDS0339/script.sql
14
old    6:      a:=&a;
new    6:      a:=14;
89
old    7:      b:=&b;
new    7:      b:=89;
B is greater

PL/SQL procedure successfully completed.

@/home/matlab/24BDS0339/script.sql
9
old    6:      a:=&a;
new    6:      a:=9;
3
old    7:      b:=&b;
new    7:      b:=3;
A is greater

PL/SQL procedure successfully completed.

SQL> 

```

```
code:
DECLARE
    msg varchar2(20);
    a number(5);
    b number(5);
begin
    a:=&a;
    b:=&b;
    if(a>b) then
        dbms_output.put_line('A is greater');
    elsif(b>a) then
        dbms_output.put_line('B is greater');
    else
        dbms_output.put_line('Both are equal');
    end if;
end;
/
```

```
@/home/matlab/24BDS0339/script.sql
5
old   6:      a:=&a;
new   6:      a:=5;
5
old   7:      b:=&b;
new   7:      b:=5;
Both are equal

PL/SQL procedure successfully completed.

SQL> □
```

```
code:
DECLARE
    e_ssn EMPLOYEE0339.ssn%type;
    e_sal EMPLOYEE0339.salary%type;
begin
    e_ssn := '&e_ssn';
    select salary into e_sal from employee0339 where ssn=e_ssn;
    dbms_output.put_line(e_ssn || ' : ' || e_sal);
end;
/
```

```
@/home/matlab/24BDS0339/script.sql
123456789
old   5:          e_ssn := '&e_ssn';
new   5:          e_ssn := '123456789';
123456789 : 30000

PL/SQL procedure successfully completed.

SQL> □
```

```

code:
DECLARE
    e_ssn EMPLOYEE0339.ssn%type;
    e_sal EMPLOYEE0339.salary%type;
begin
    e_ssn := '&e_ssn';
    select salary into e_sal from employee0339 where ssn=e_ssn;
    dbms_output.put_line(e_ssn || ' : ' || e_sal);

    if(e_sal < 30000) then
        update employee0339 set salary = e_sal+5000 where ssn =
e_ssn;
    else
        update employee0339 set salary = e_sal+2000 where ssn =
e_ssn;
    end if;
end;
/

```

```

select ssn, salary from employee0339;

SSN          SALARY
-----
554433221    80000
543216789    70000
333445555    40000
987654321    43000
123456789    30000
666884444    38000
453453453    25000
888665555    55000
999887777    25000
987987987    25000
943775543    58000

11 rows selected.

@/home/matlab/24BDS0339/script.sql
123456789
old 5:      e_ssn := '&e_ssn';
new 5:      e_ssn := '123456789';
123456789 : 30000

PL/SQL procedure successfully completed.

select ssn, salary from employee0339;

SSN          SALARY
-----
554433221    80000
543216789    70000
333445555    40000
987654321    43000
123456789    32000
666884444    38000
453453453    25000
888665555    55000
999887777    25000
987987987    25000
943775543    58000

11 rows selected.

SQL> 

```



```
code:
DECLARE
    day number(5);
begin
    day := &day;

    case day
        when 1 then
            dbms_output.put_line('Monday');
        when 2 then
            dbms_output.put_line('Tuesday');
        else
            dbms_output.put_line('Invalid');
        end case;
end;
/
```

```
@/home/matlab/24BDS0339/script.sql
1
old 4:      day := &day;
new 4:      day := 1;
Monday

PL/SQL procedure successfully completed.

@/home/matlab/24BDS0339/script.sql
2
old 4:      day := &day;
new 4:      day := 2;
Tuesday

PL/SQL procedure successfully completed.

@/home/matlab/24BDS0339/script.sql
6
old 4:      day := &day;
new 4:      day := 6;
Invalid

PL/SQL procedure successfully completed.

SQL> █
```

```

code:
DECLARE
    mark number(3);
begin
    mark := &mark;

    case
        when mark>90 then
            dbms_output.put_line('S Grade');
        when mark>80 then
            dbms_output.put_line('A Grade');
        when mark>50 then
            dbms_output.put_line('F Grade');
        else
            dbms_output.put_line('Invalid');
    end case;
end;

```

```

@/home/matlab/24BDS0339/script.sql
95
old 4:      mark := &mark;
new 4:      mark := 95;
S Grade

PL/SQL procedure successfully completed.

@/home/matlab/24BDS0339/script.sql
87
old 4:      mark := &mark;
new 4:      mark := 87;
A Grade

PL/SQL procedure successfully completed.

@/home/matlab/24BDS0339/script.sql
55
old 4:      mark := &mark;
new 4:      mark := 55;
F Grade

PL/SQL procedure successfully completed.

@/home/matlab/24BDS0339/script.sql
49
old 4:      mark := &mark;
new 4:      mark := 49;
Invalid

PL/SQL procedure successfully completed.

SQL> 

```

```
code:
DECLARE
    x number(3) := 10;
begin
    loop
        dbms_output.put_line(x);
        x:=x+10;
        if (x>50) then
            exit;
        end if;
    end loop;
end;
/
```

```
@/home/matlab/24BDS0339/script.sql
10
20
30
40
50

PL/SQL procedure successfully completed.
SQL> □
```

```

code:
DECLARE
    x number(3) := 10;
begin
    loop
        dbms_output.put_line(x);
        x:=x+10;
        if (x>50) then
            exit;
        end if;
    end loop;

    for x in 1 .. 10 loop
        dbms_output.put_line(x);
    end loop;

    for x in reverse 1 .. 10 loop
        dbms_output.put_line(x || ' ');
    end loop;
end;
/

```

```

@/home/matlab/24BDS0339/script.sql
10
20
30
40
50
1
2
3
4
5
6
7
8
9
10
10
9
8
7
6
5
4
3
2
1
PL/SQL procedure successfully completed.
SQL> 

```

```
code:
DECLARE
    x number(3) := 10;
begin
    loop
        dbms_output.put_line(x);
        x:=x+10;
        if (x>50) then
            exit;
        end if;
    end loop;

    for x in 1 .. 10 loop
        dbms_output.put_line(x);
    end loop;

    for x in reverse 1 .. 10 loop
        dbms_output.put_line(x || ' ');
    end loop;

    x:=10;

    while x>0 loop
        dbms_output.put_line(x);
        x:= x-1;
    end loop;
end;
/
```

```
@/home/matlab/24BDS0339/script.sql
```

```
10
```

```
20
```

```
30
```

```
40
```

```
50
```

```
1
```

```
2
```

```
3
```

```
4
```

```
5
```

```
6
```

```
7
```

```
8
```

```
9
```

```
10
```

```
10
```

```
9
```

```
8
```

```
7
```

```
6
```

```
5
```

```
4
```

```
3
```

```
2
```

```
1
```

```
10
```

```
9
```

```
8
```

```
7
```

```
6
```

```
5
```

```
4
```

```
3
```

```
2
```

```
1
```

```
PL/SQL procedure successfully completed.
```

```
SQL> □
```